R125/code_seed2 | Single VAL-best full18 results

Frozen new90k, BQ/LEHD deleted; FP64 reduced-pool argmin; no TEST/LIB reselection.

epoch 28 | update 40320 | seed2 | fresh initialization

Data: selection_data_new90k_no_bq_lehd_seed2_20261112

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	73.1062	68.7198	11.417483	11.427978	11.376374	-0.1813	+0.3644	0.3689
TEST ALL	73.2617	68.6481	11.374641	11.385299	11.334478	-0.1862	+0.3564	0.3603

Dataset	n	Fixed NSS Top1 (%)	R125 Top1 (%)	Fixed NSS cost	R125 cost
TSP TEST	4500	68.3111	68.4667	7.98877858	7.98872225
CVRP TEST	4500	61.5111	62.1778	18.00658479	18.00611219
TSPLIB	49	14.2857	22.4490	8.23925650	8.22452230
CVRPLIB	100	71.0000	63.0000	67.10111172	67.29271499

NSS: fixed historical logits reselected after deleting BQ/LEHD; NOT a matched reduced-pool training control.

Only existing aligned4500-row TEST / original LIB predictions are reused; unavailable entries are MISSING.

SBS: reduced-pool full81000 TRAIN only. Oracle: minimum legal FP64 cost per instance.

ALL: equal weight per problem. Macro raw costs are descriptive, not cross-problem pooled costs.

TRAIN81000 / VAL4500 / TEST4500 per problem. Retained holdouts informed earlier R&D, not fully blind.

R80D: architecture reference only; older data / original pool, NOT a matched experimental control.

Deletion changes Oracle and winner labels; a Top1 change alone does not establish an architectural cause.

R125/code_seed2 | VAL: ALL and18 tasks

Top1 higher is better; cost/regret lower is better. Same checkpoint for every task.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	73.1062	68.7198	11.417483	11.427978	11.376374	-0.1813	+0.3644	0.3689	per-problem TRAIN SBS
TSP	68.7111	66.5556	8.044430	8.045354	8.031828	-0.0115	+0.1569	0.1690	T2T
CVRP	62.0667	30.5556	18.151493	18.269377	18.107061	-0.6453	+0.2454	0.2925	RELD_CVRP
ATSP	77.2444	47.4889	1.638661	1.676905	1.632471	-2.2806	+0.3792	0.3753	UNICO_MatPOENet
OVRP	40.2667	35.7556	7.462608	7.469579	7.415777	-0.0933	+0.6315	0.6461	MoSES_RF
VRPB	80.9778	80.8444	9.354257	9.354876	9.318750	-0.0066	+0.3810	0.3905	reld_mt
VRPL	38.4444	33.0889	12.432011	12.437959	12.383444	-0.0478	+0.3922	0.4009	MoSES_RF
VRPTW	80.0667	79.9556	18.818914	18.819431	18.766139	-0.0027	+0.2812	0.2970	reld_mt
OVRPTW	83.8444	83.7778	10.143288	10.143730	10.116859	-0.0044	+0.2612	0.2581	reld_mt
OVRPB	88.4889	88.5111	6.357290	6.357254	6.341078	+0.0006	+0.2557	0.2656	reld_mt
OVRPL	40.4667	35.9111	7.439096	7.445709	7.390416	-0.0888	+0.6587	0.6785	MoSES_RF
VRPBL	78.8222	78.5111	9.368383	9.371523	9.328366	-0.0335	+0.4290	0.4317	reld_mt
VRPBTW	78.5556	78.4222	19.947404	19.950430	19.851084	-0.0152	+0.4852	0.4650	reld_mt
VRPLTW	78.8222	78.7556	18.764463	18.765871	18.707843	-0.0075	+0.3027	0.3187	reld_mt
OVRPBL	89.3333	89.2000	6.369305	6.370093	6.352824	-0.0124	+0.2594	0.2695	reld_mt
OVRPBTW	83.5333	83.4444	10.660678	10.661098	10.623245	-0.0039	+0.3524	0.3253	reld_mt
OVRPLTW	83.5556	83.4889	10.157069	10.157423	10.129818	-0.0035	+0.2690	0.2715	reld_mt
VRPBLTW	79.4889	79.4000	19.794949	19.797044	19.706420	-0.0106	+0.4492	0.4376	reld_mt
OVRPBLTW	83.2222	83.2889	10.610392	10.609955	10.571302	+0.0041	+0.3698	0.3470	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; NOT a ratio of means.

Native FP64 argmin after candidate deletion; never FP32 cost argmin.

R125/code_seed2 | TEST: ALL and18 tasks

Top1 higher is better; cost/regret lower is better. Same checkpoint for every task.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	73.2617	68.6481	11.374641	11.385299	11.334478	-0.1862	+0.3564	0.3603	per-problem TRAIN SBS
TSP	68.4667	67.1333	7.988722	7.989621	7.976503	-0.0113	+0.1532	0.1669	T2T
CVRP	62.1778	29.7556	18.006112	18.124983	17.961416	-0.6558	+0.2488	0.2977	RELD_CVRP
ATSP	76.9111	47.2444	1.635400	1.674207	1.629441	-2.3180	+0.3657	0.3657	UNICO_MatPOENet
OVRP	40.4222	33.6444	7.446207	7.455750	7.400487	-0.1280	+0.6178	0.6288	MoSES_RF
VRPB	80.5556	80.4000	9.350323	9.351810	9.315298	-0.0159	+0.3760	0.3794	reld_mt
VRPL	37.4222	32.2667	12.350021	12.356138	12.300958	-0.0495	+0.3989	0.4083	MoSES_RF
VRPTW	81.0667	80.8444	18.690011	18.690545	18.637408	-0.0029	+0.2822	0.2937	reld_mt
OVRPTW	83.9556	83.9333	10.092720	10.093063	10.068135	-0.0034	+0.2442	0.2353	reld_mt
OVRPB	88.0889	88.1333	6.370906	6.370826	6.352225	+0.0013	+0.2941	0.3020	reld_mt
OVRPL	41.5111	34.6000	7.463809	7.472542	7.418033	-0.1169	+0.6171	0.6307	MoSES_RF
VRPBL	79.6000	79.4222	9.394514	9.396379	9.355967	-0.0199	+0.4120	0.4110	reld_mt
VRPBTW	78.8000	78.7556	19.665943	19.665725	19.577666	+0.0011	+0.4509	0.4493	reld_mt
VRPLTW	79.2444	79.2444	18.802013	18.803136	18.745143	-0.0060	+0.3034	0.3209	reld_mt
OVRPBL	89.3111	89.2889	6.375617	6.375741	6.359310	-0.0019	+0.2564	0.2675	reld_mt
OVRPBTW	84.9333	84.8222	10.658850	10.659591	10.624748	-0.0070	+0.3210	0.2934	reld_mt
OVRPLTW	82.9333	82.8667	10.085820	10.086112	10.059001	-0.0029	+0.2666	0.2670	reld_mt
VRPBLTW	79.6667	79.7333	19.704770	19.707068	19.613284	-0.0117	+0.4664	0.4501	reld_mt
OVRPBLTW	83.6444	83.5778	10.661784	10.662149	10.625574	-0.0034	+0.3408	0.3170	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; NOT a ratio of means.

Native FP64 argmin after candidate deletion; never FP32 cost argmin.

R125/code_seed2 | Native LIB and provenance

Original LIB inputs/costs; BQ/LEHD removed; winner recomputed in FP64; no new solver answers.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	22.4490	12.2449	8.224522	8.240435	8.171164	-0.1931	+0.6530	0.6269
CVRPLIB	63.0000	48.0000	67.292715	67.245364	66.961613	+0.0704	+0.4945	0.5563

SHA256: 38c4b4c540a9e352294d2bc1439ea74137fa140d3d44e913b5b6bb52f3264bf0

Selection: strict minimum full18 VAL macro_vs_sbs_pct; reduced-pool full TRAIN SBS; first wins ties

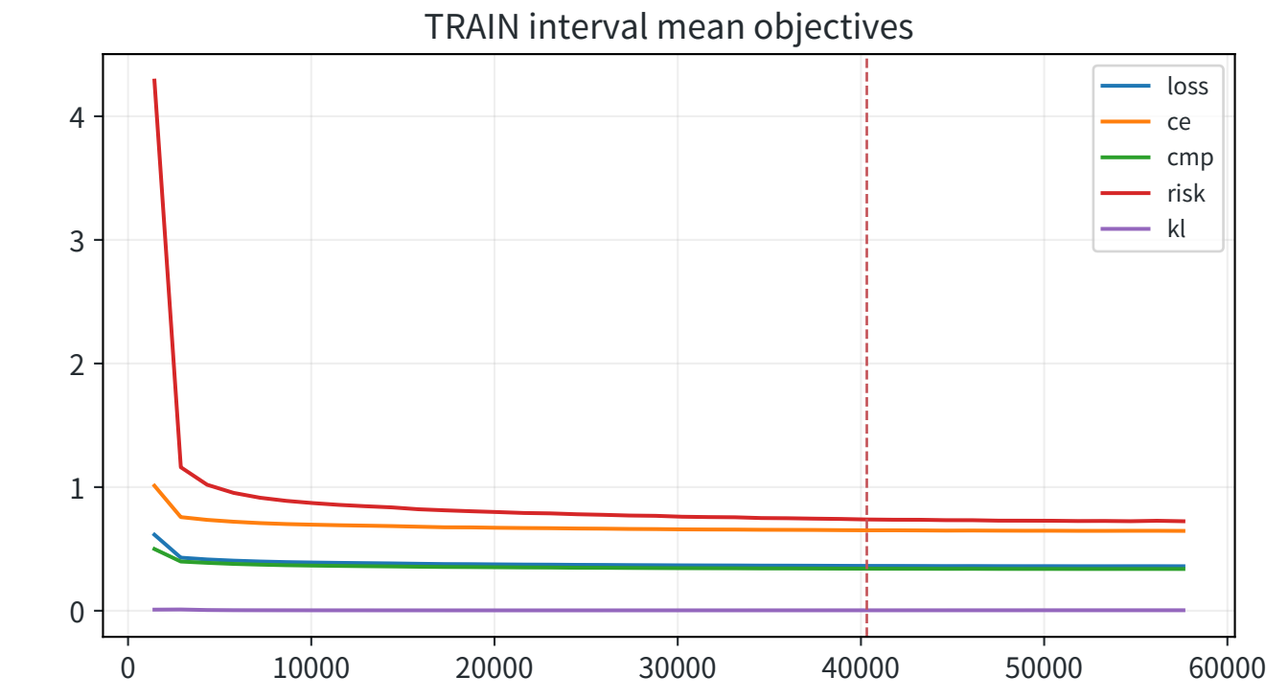
Factory: code.V4.r125_experiment.load_checkpoint

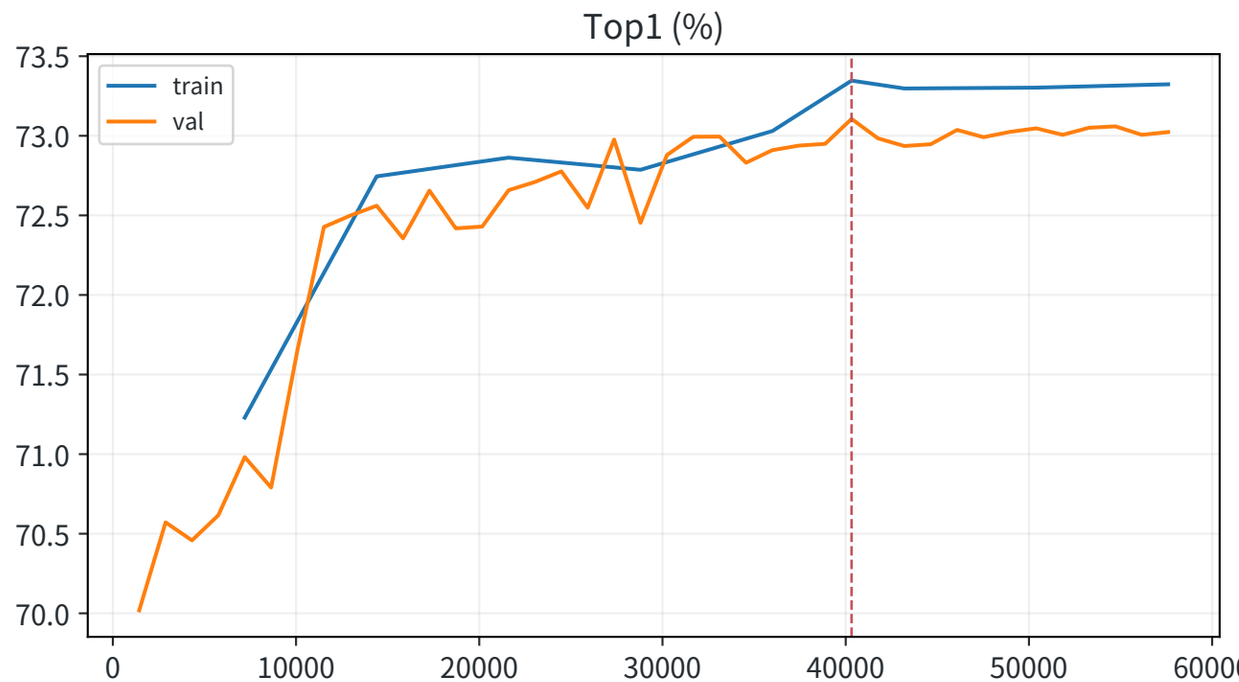
Inference: original utility score differences and maximin; reduced candidates; ascending-global-ID tie break

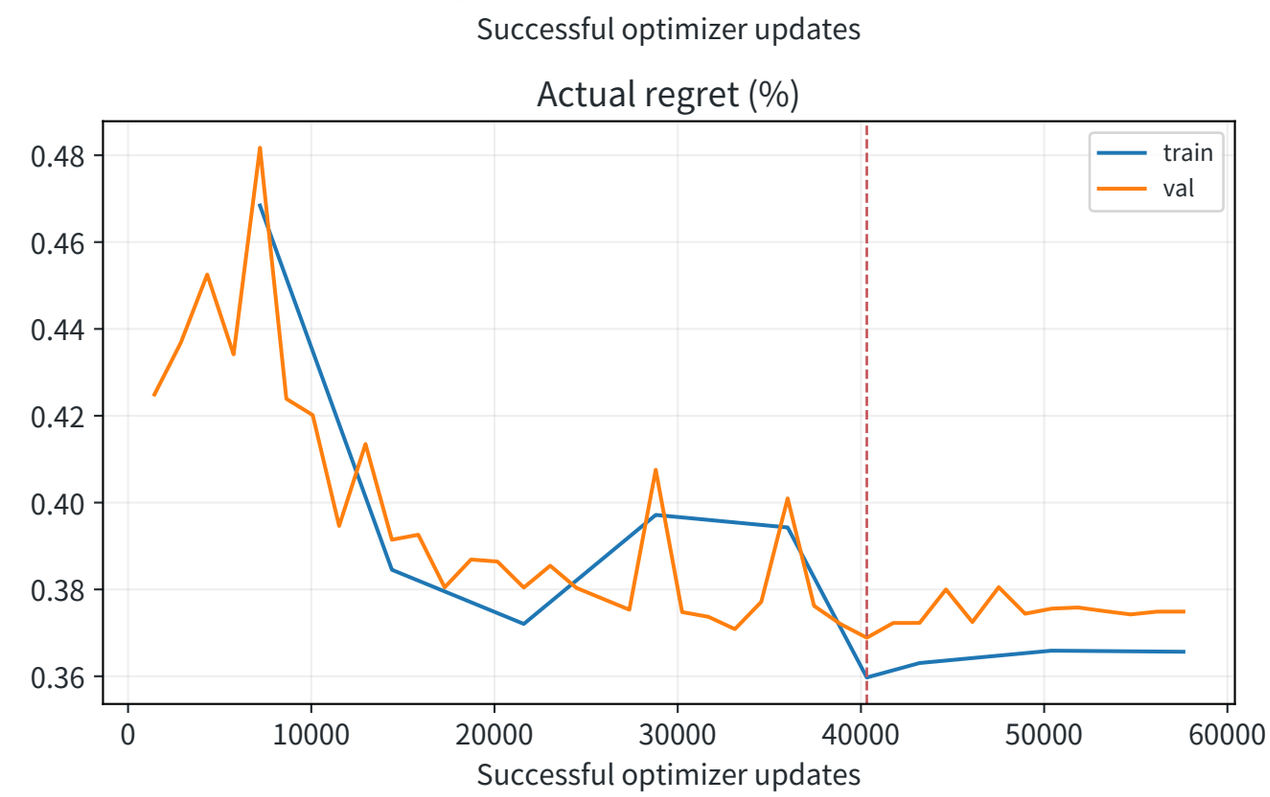
TSP6 / CVRP8; other16 task pools unchanged.

TRAIN-only normalization restored from the same best.pt; no LIB refit.

R125/code_seed2 | one locked VAL checkpoint







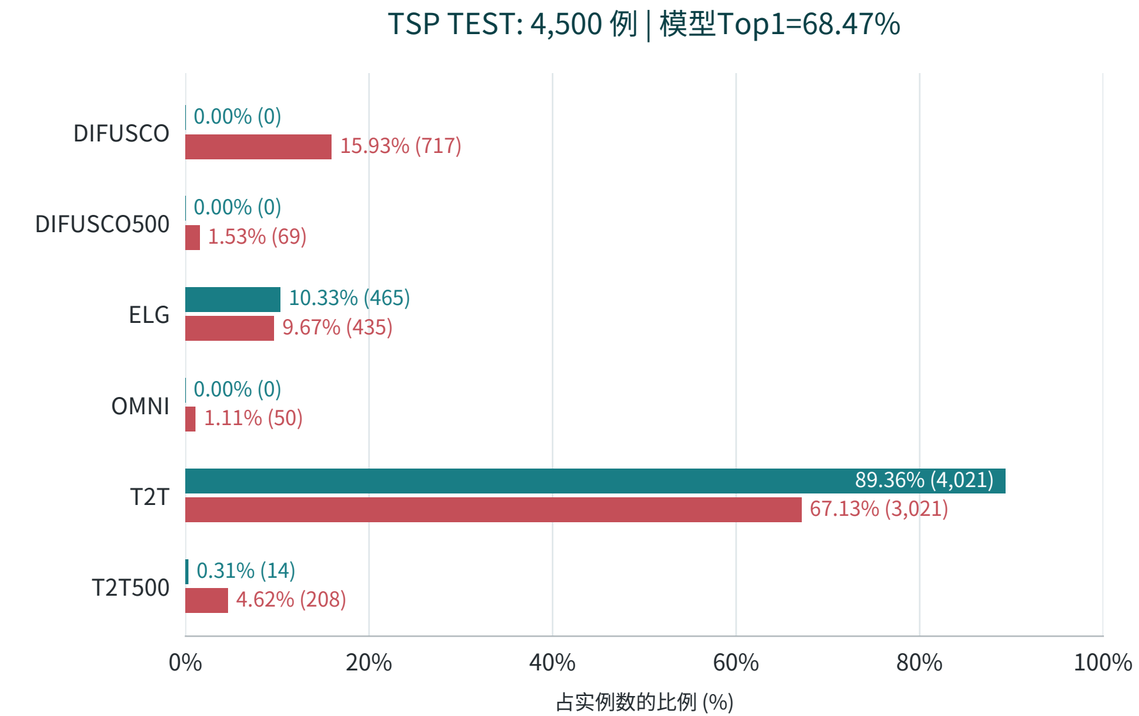


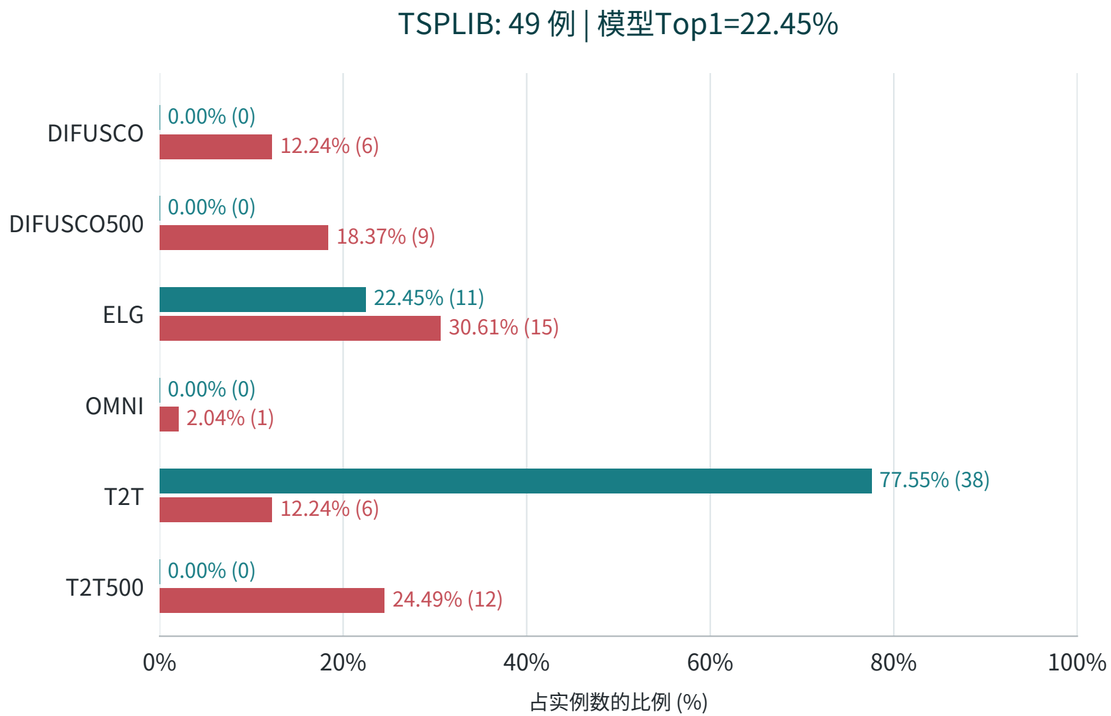
R125/code_seed2 | TSP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

LIB: data/TSPLIB | FP64 argmin after BQ/LEHD deletion





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

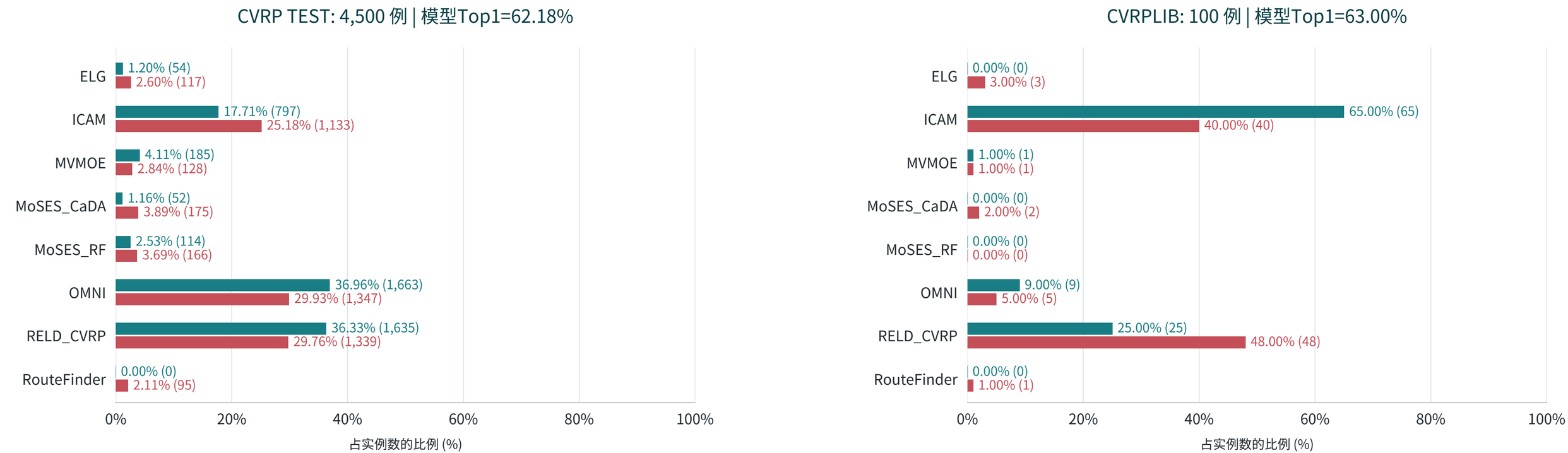
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | CVRP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

LIB: data/CVRPLIB | FP64 argmin after BQ/LEHD deletion



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

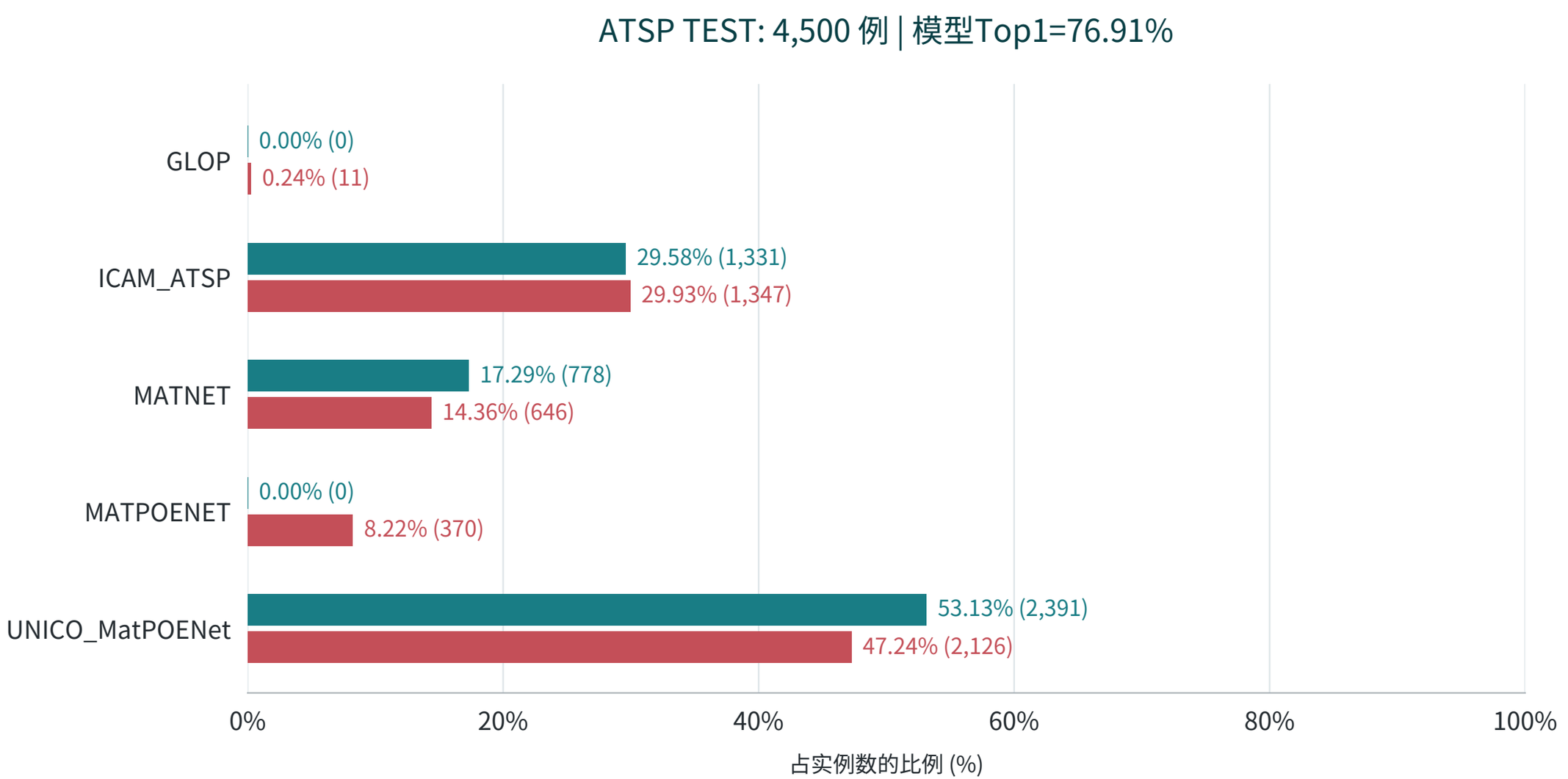
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | ATSP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

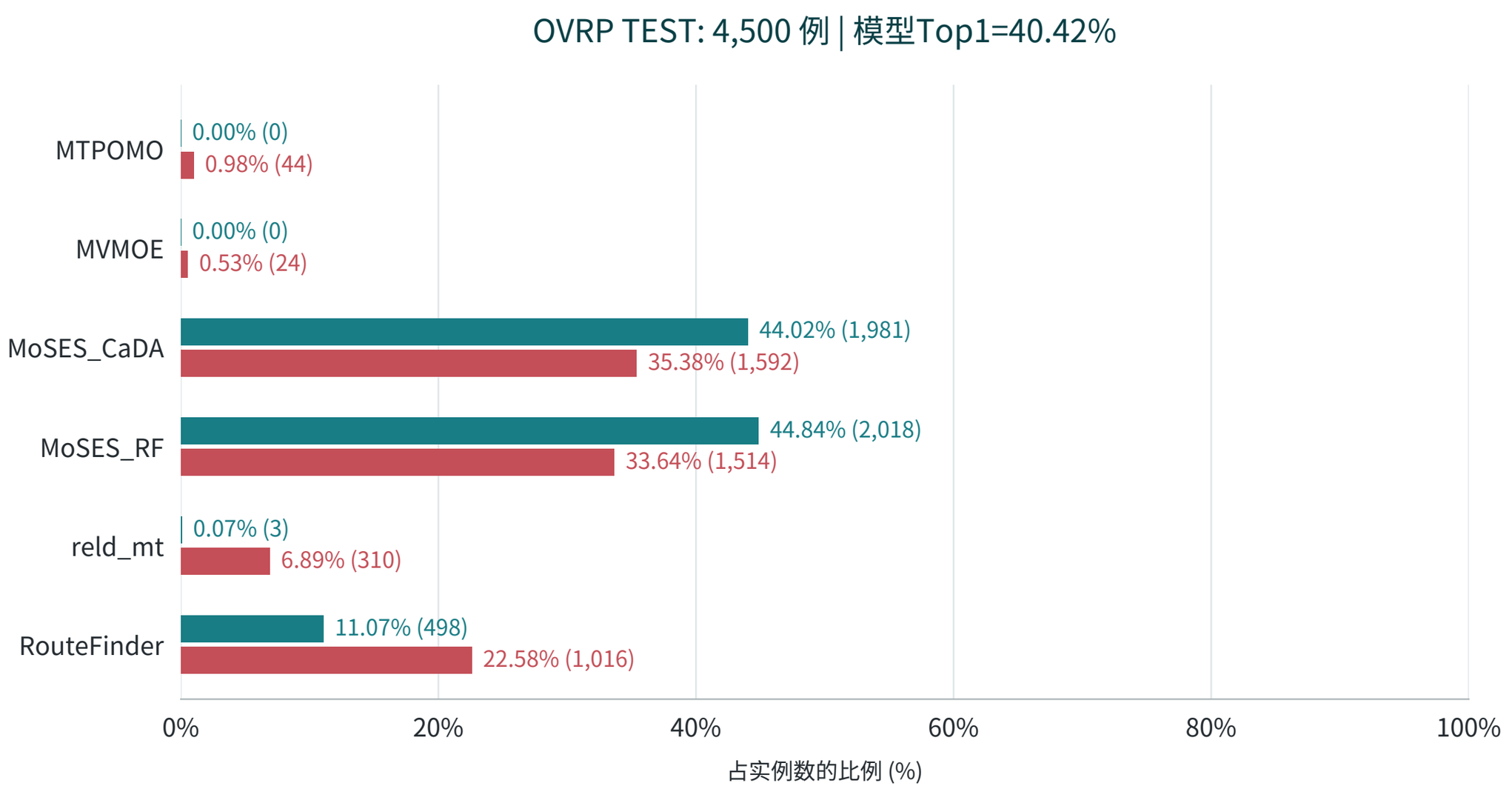
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRP | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

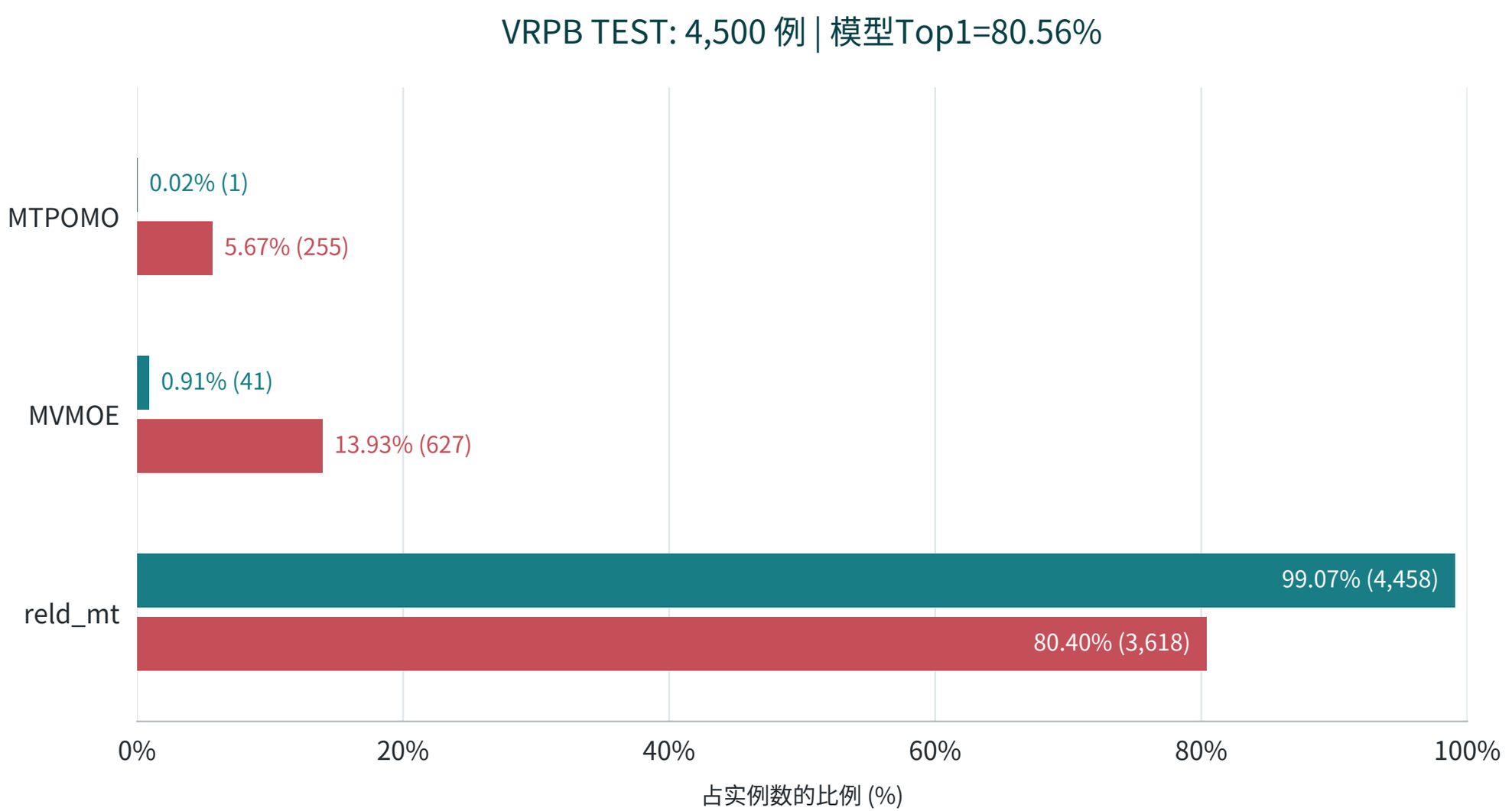
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPB | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

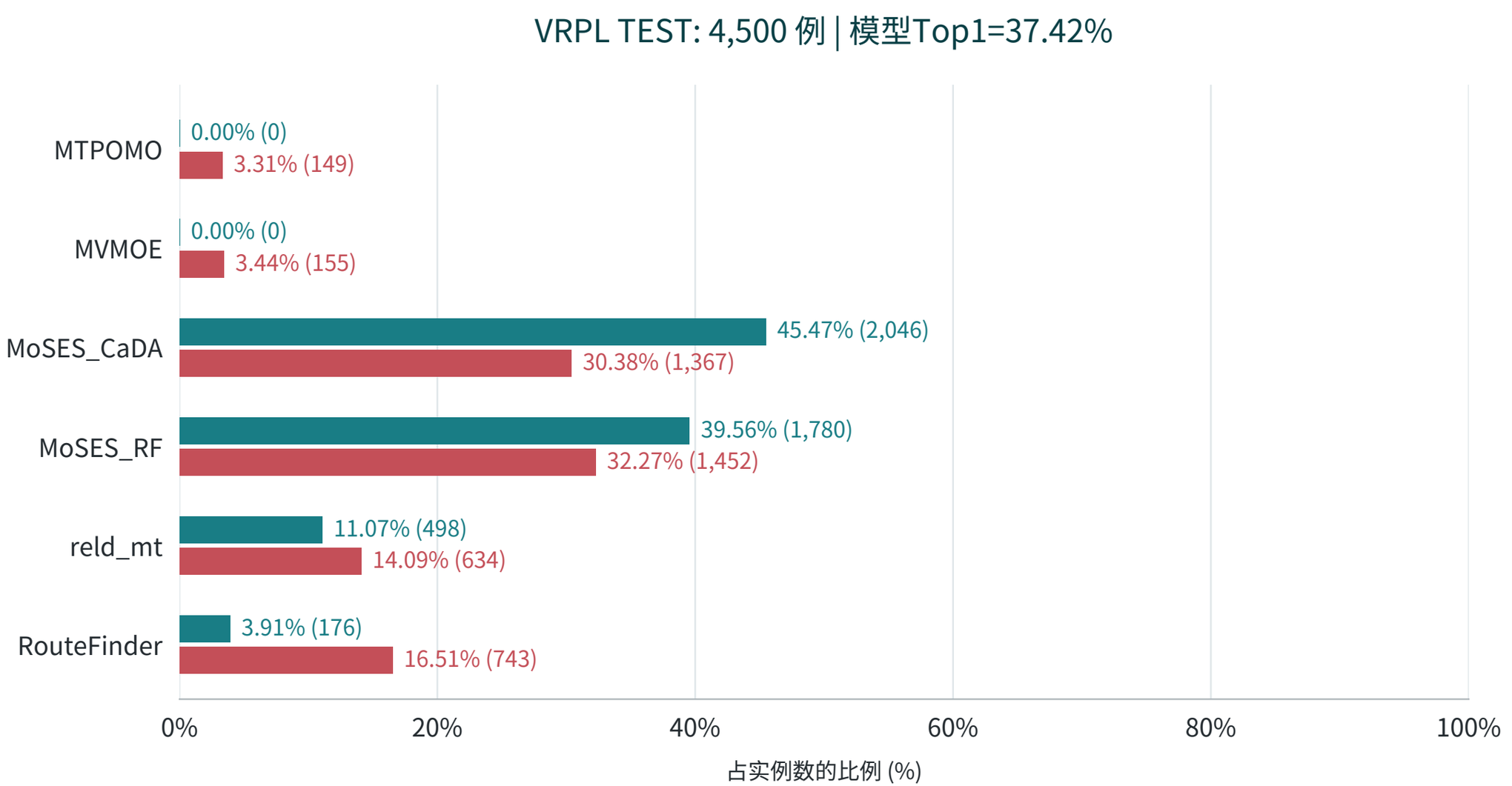
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

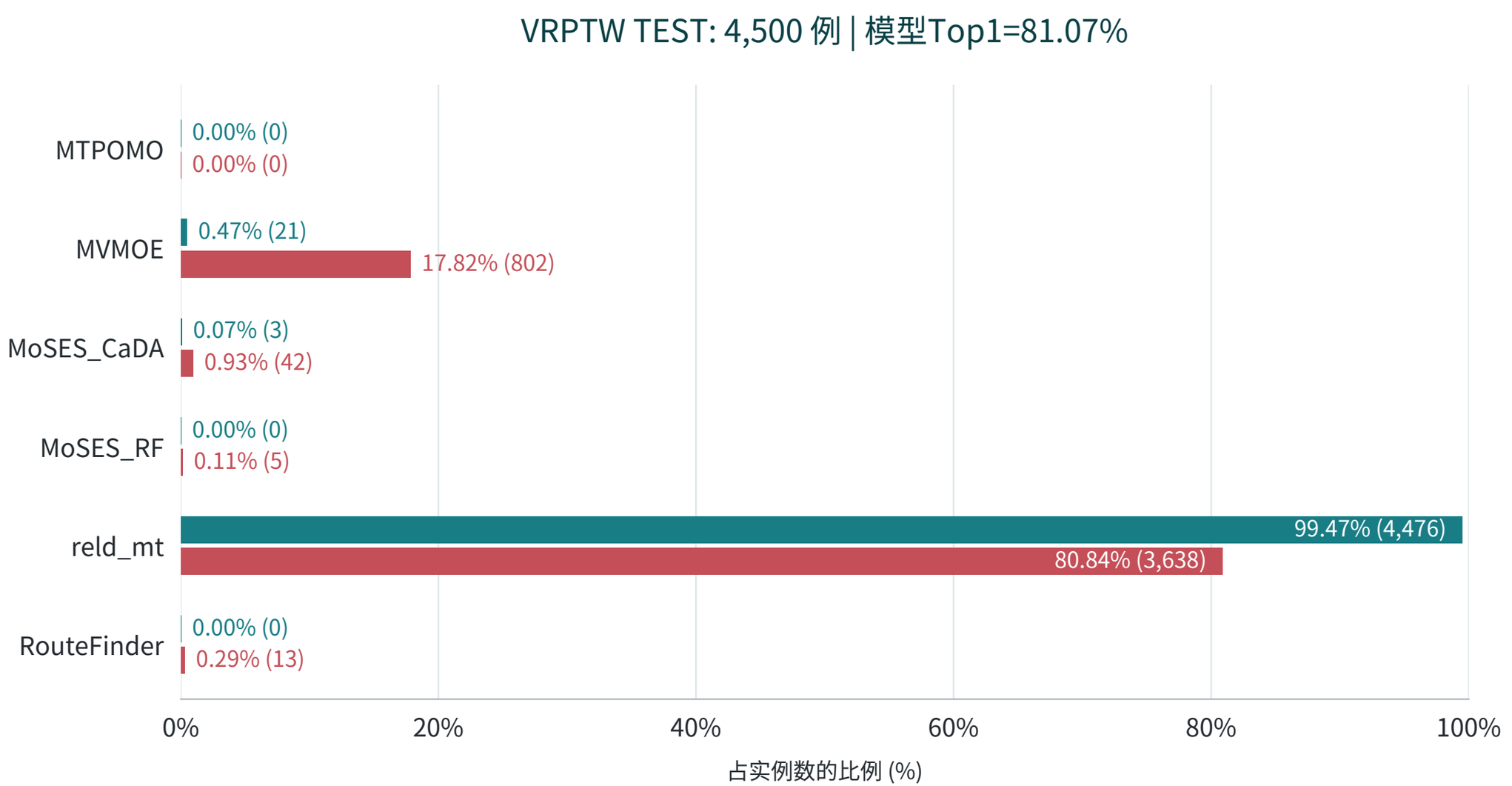
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

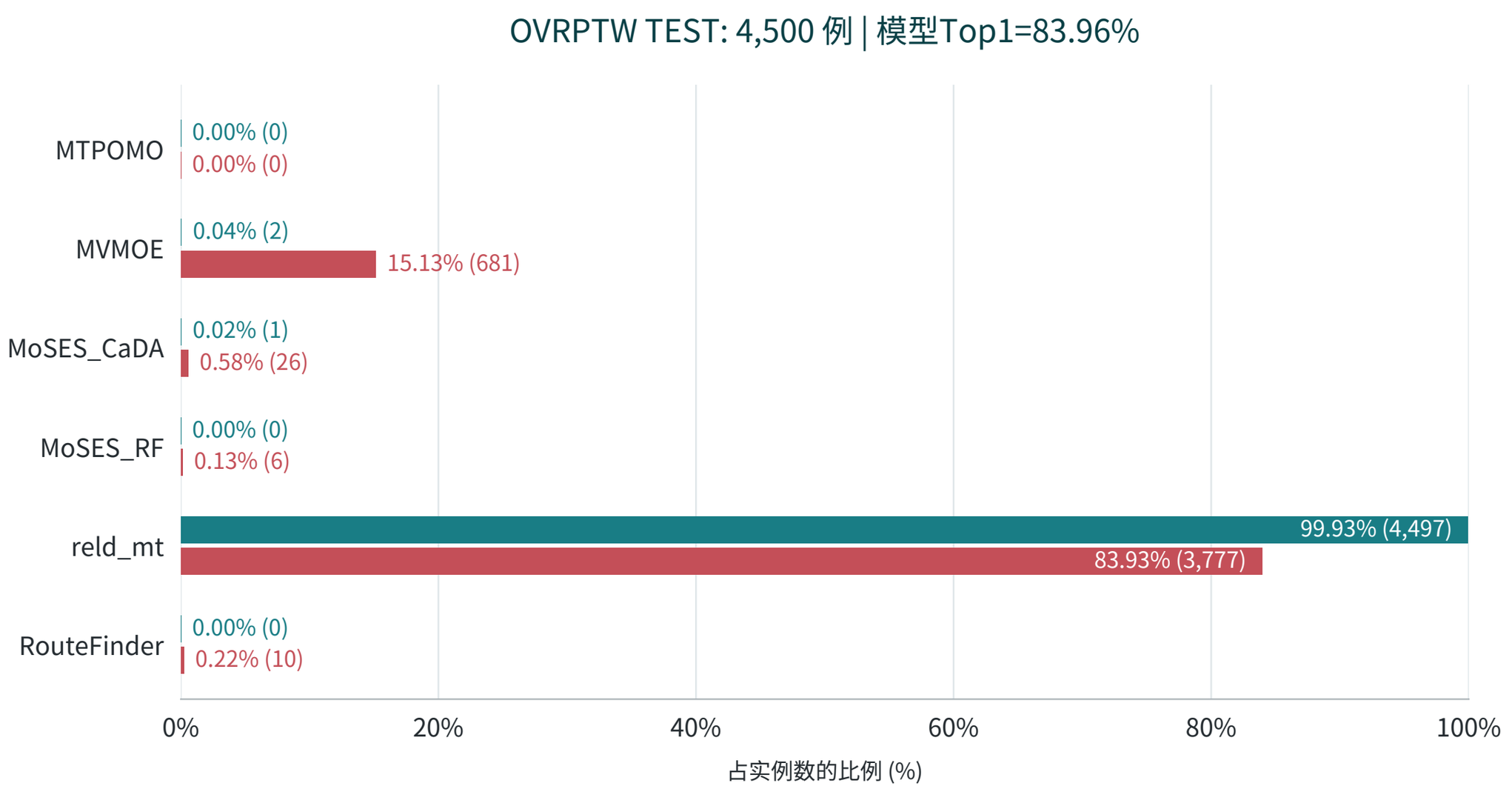
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

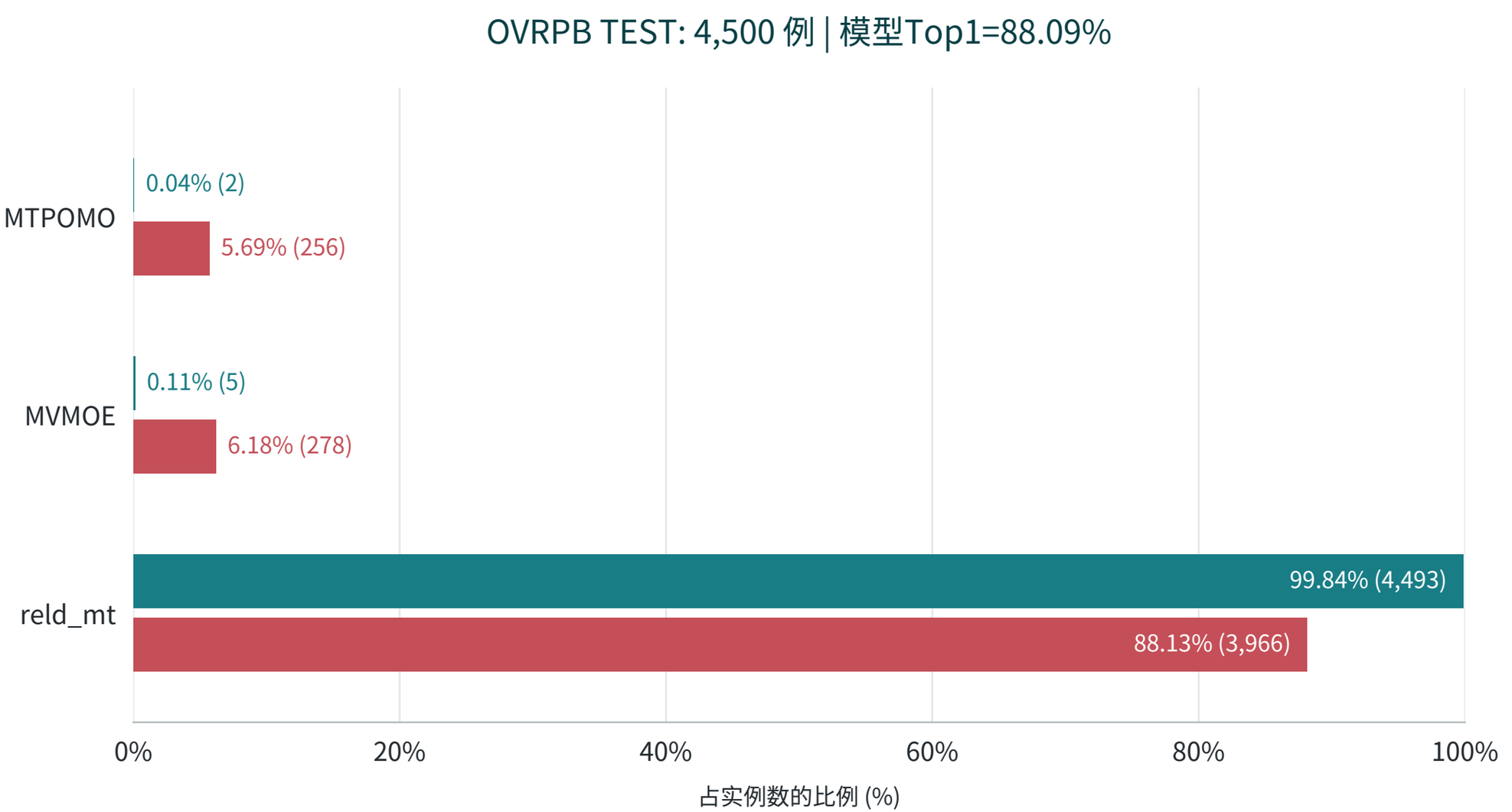
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPB | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

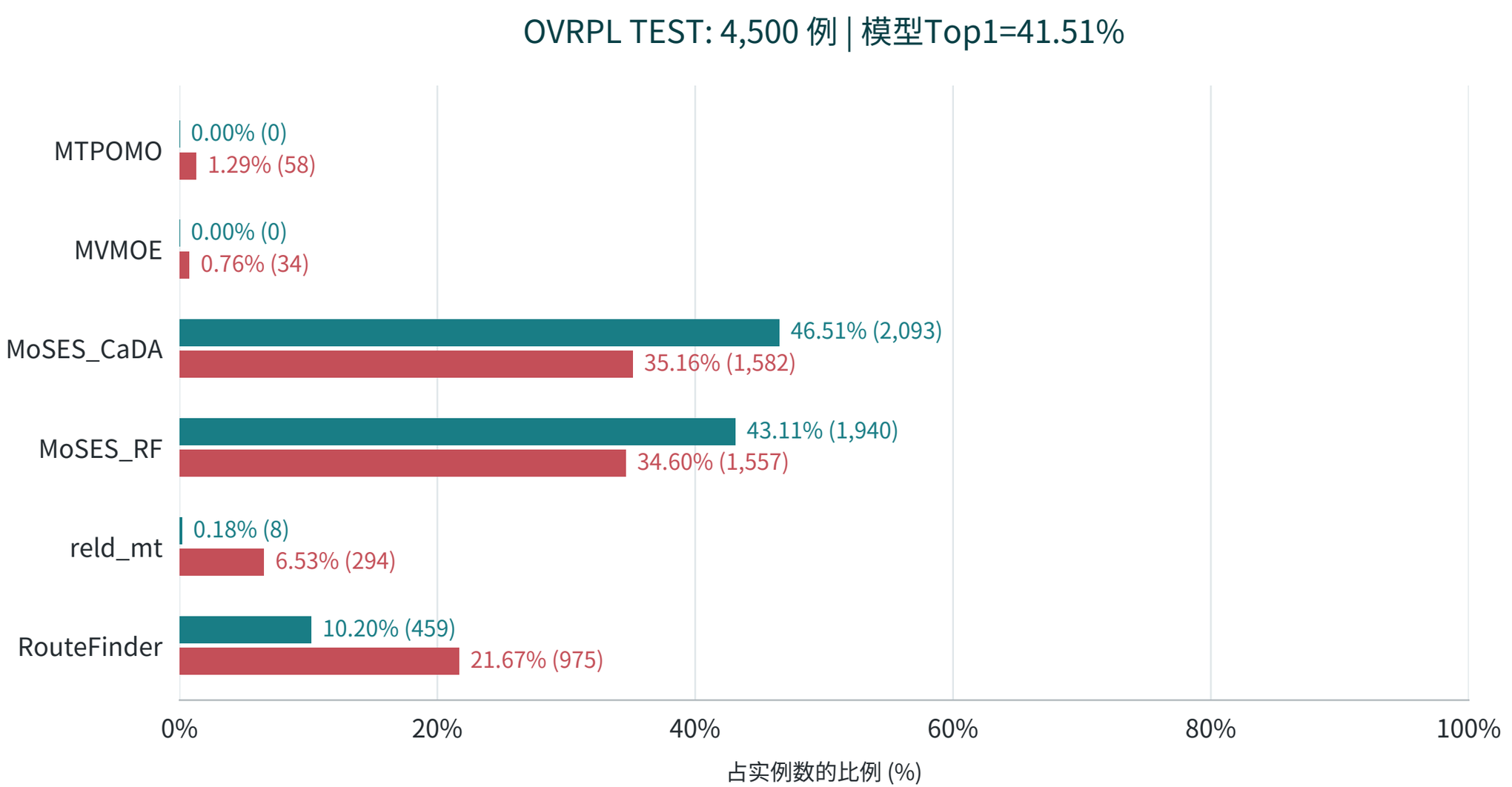
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

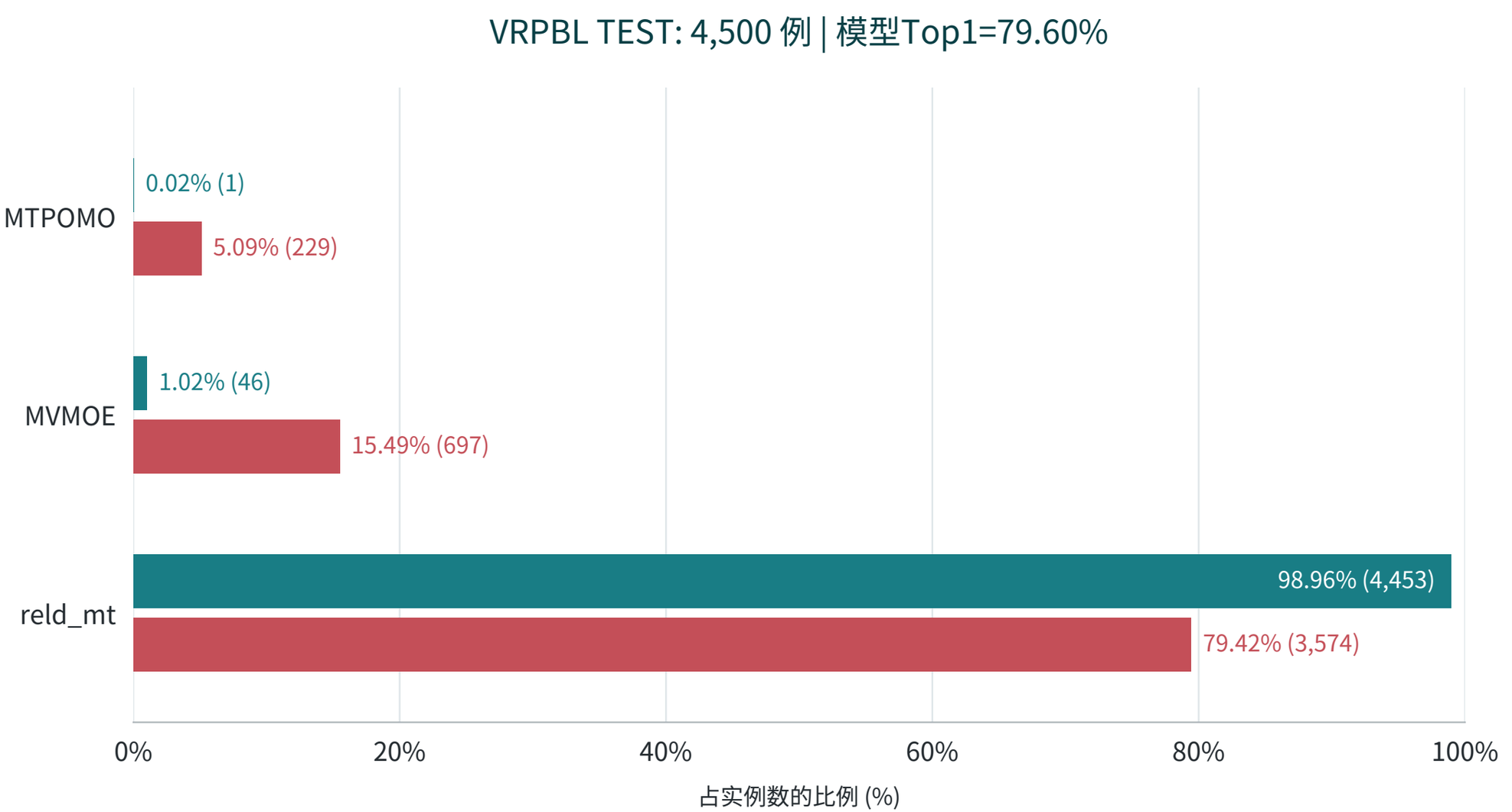
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPBL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

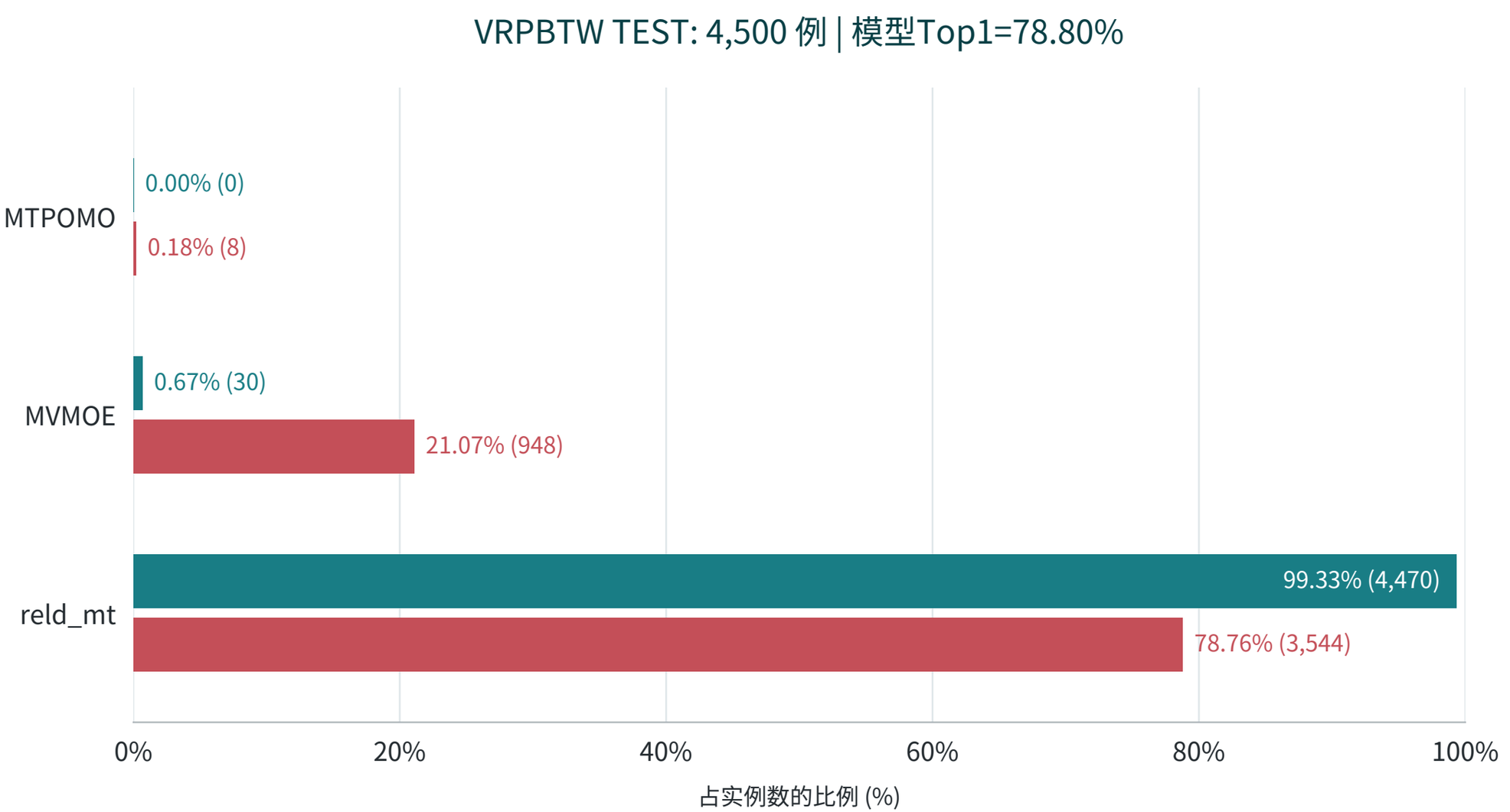
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPBTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

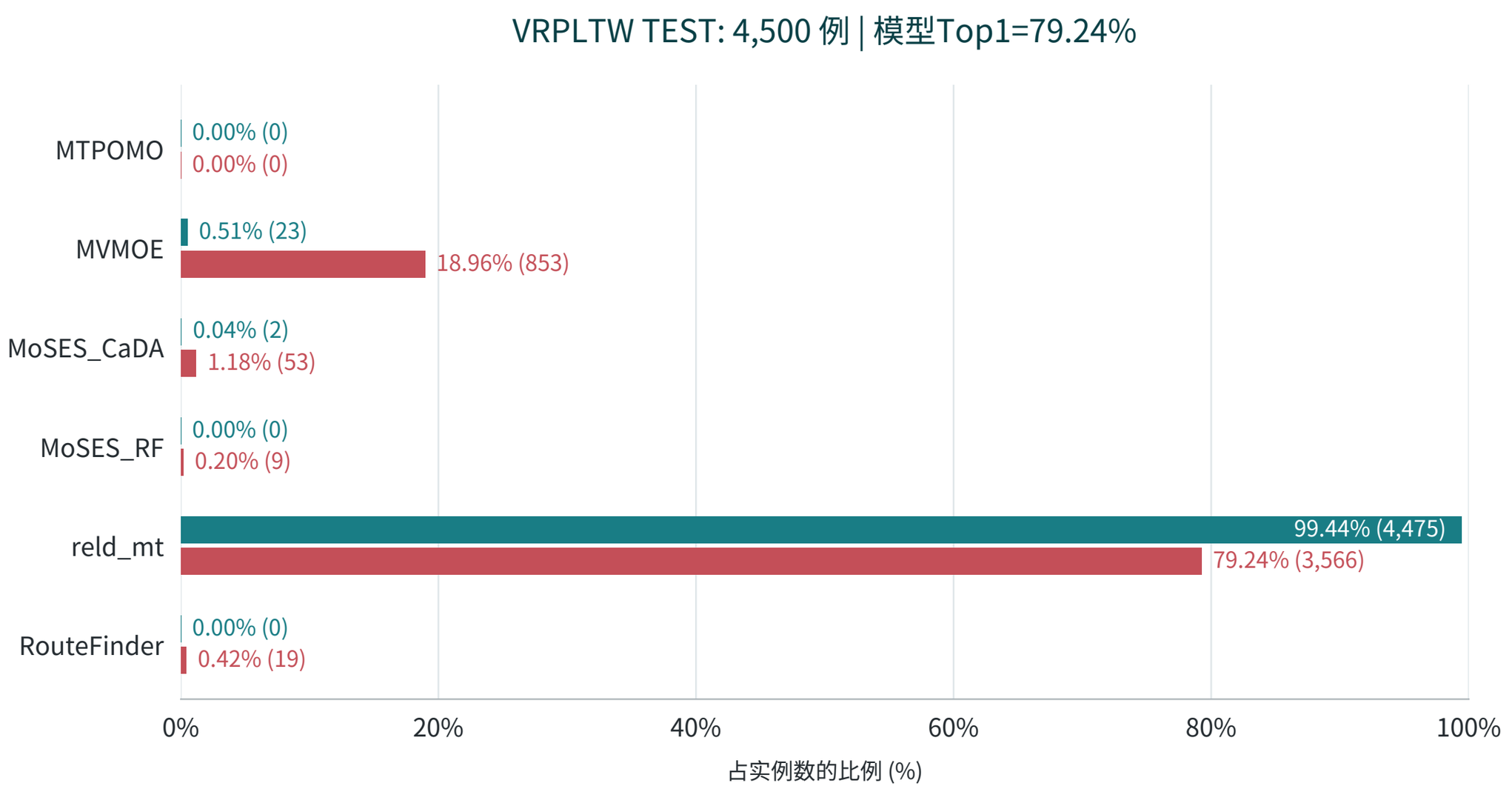
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPLTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

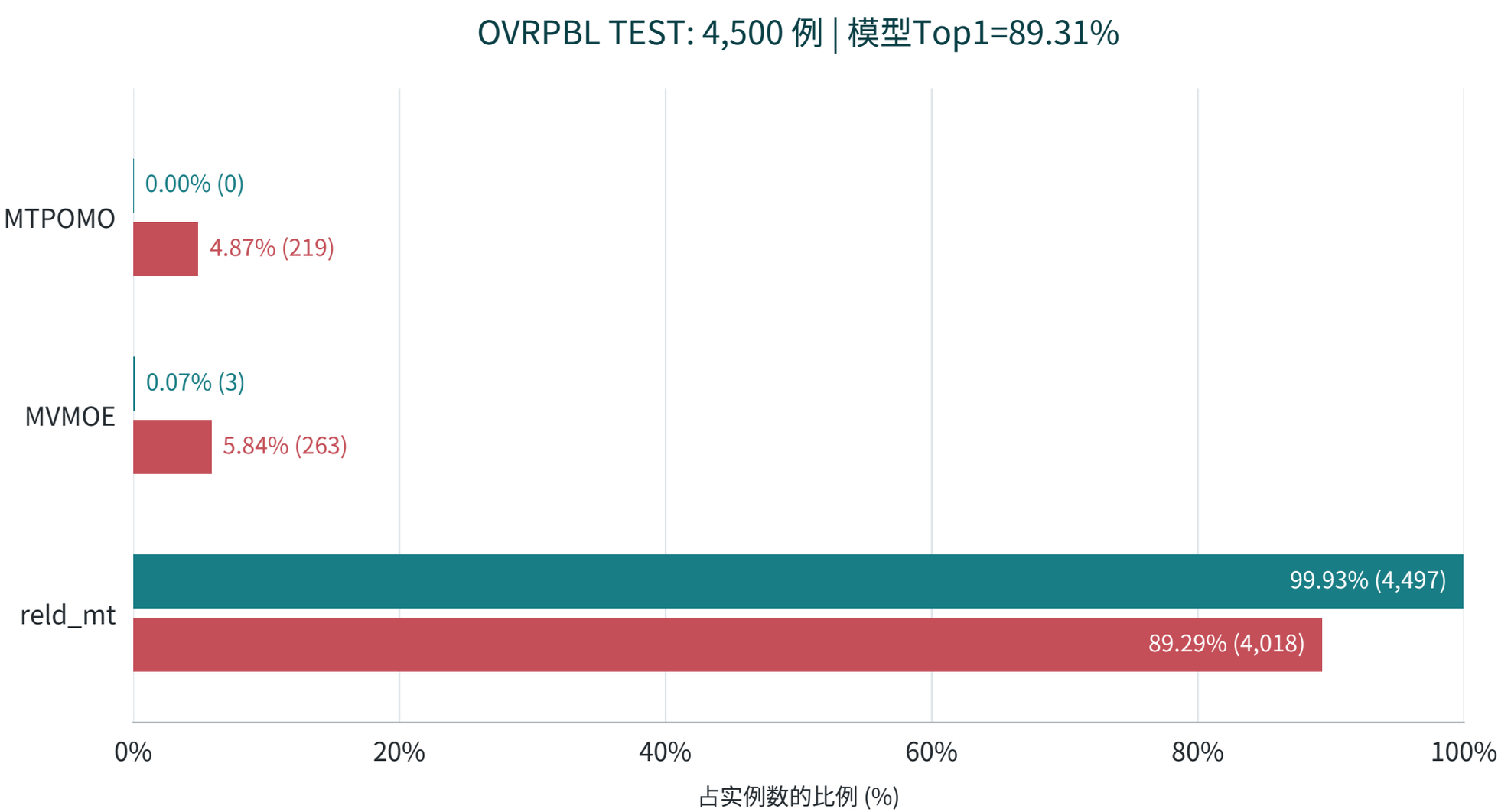
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPBL | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

新 TEST 数据版本: selection_data_new90k_no_bq_lehd_seed2_20261112

标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

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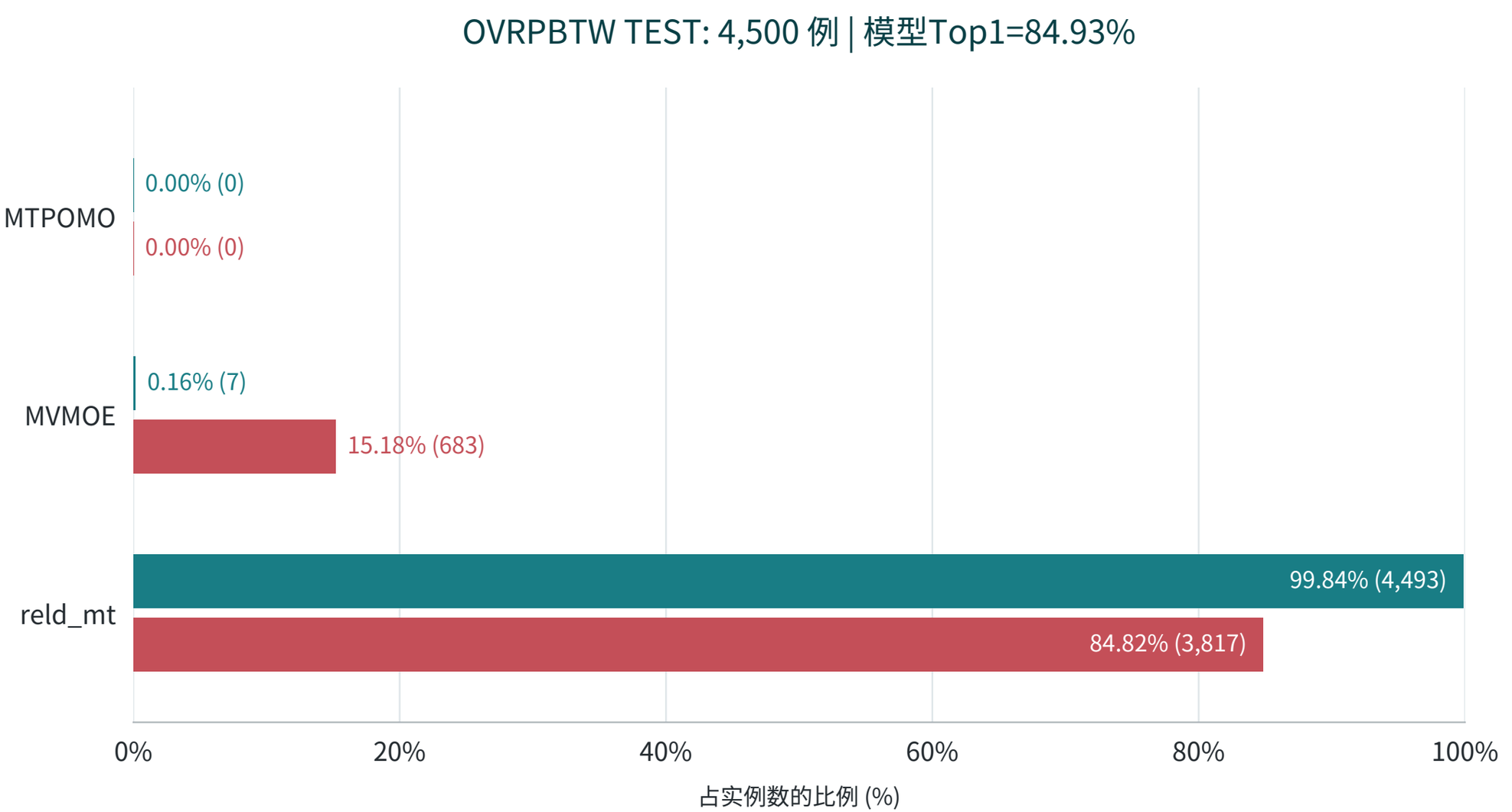
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPBTW | 方法分布

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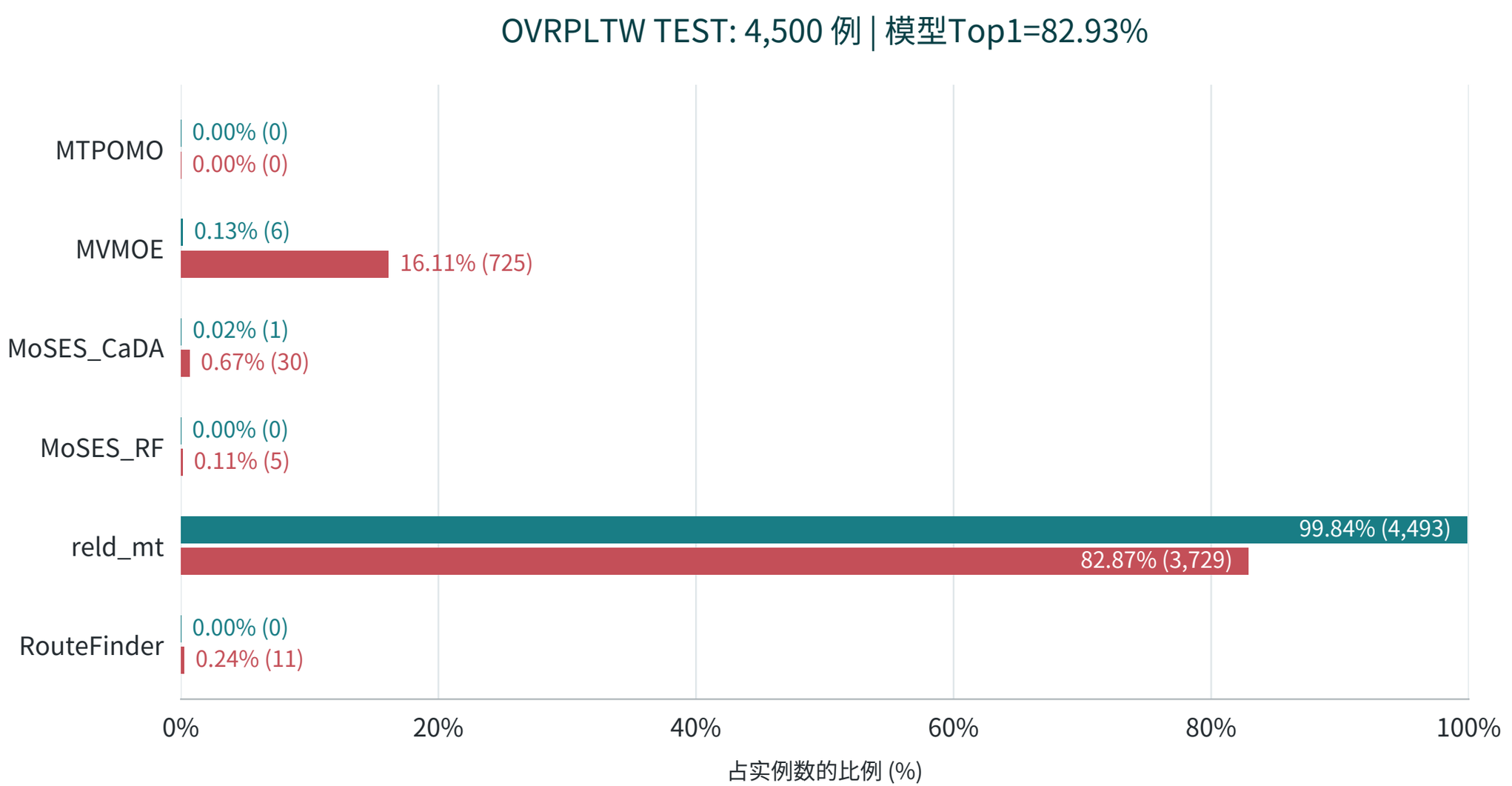
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | OVRPLTW | 方法分布

同一锁定模型: epoch 28 / update 40320 / SHA256 38c4b4c540a9e352

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标签: 原生 FP64 argmin after BQ/LEHD deletion; 仅读锁定预测



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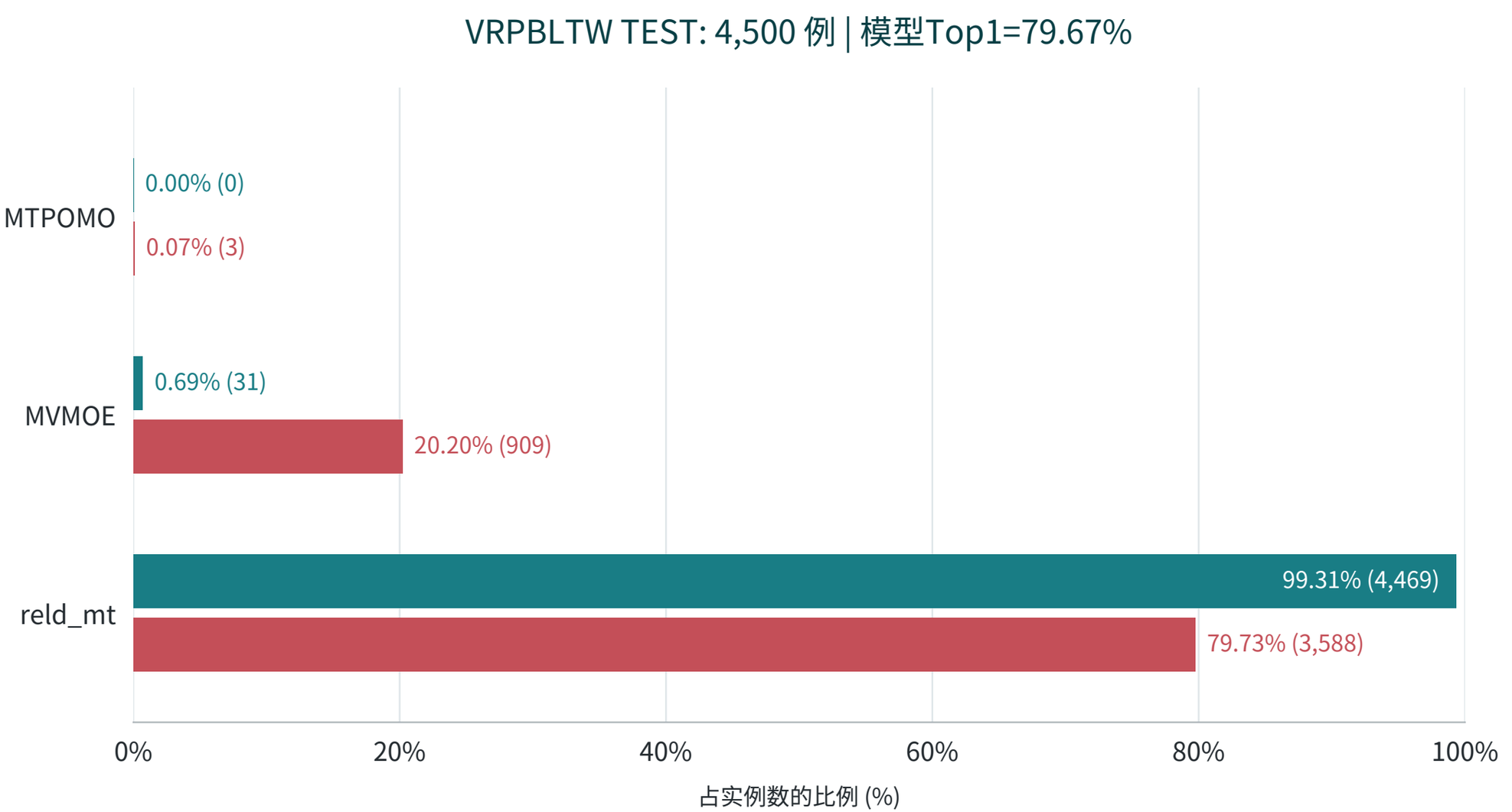
R125: BQ/LEHD deleted; same VAL lock; ascending global-ID score ties; no Rxx mixing.

R125/code_seed2 | VRPBLTW | 方法分布

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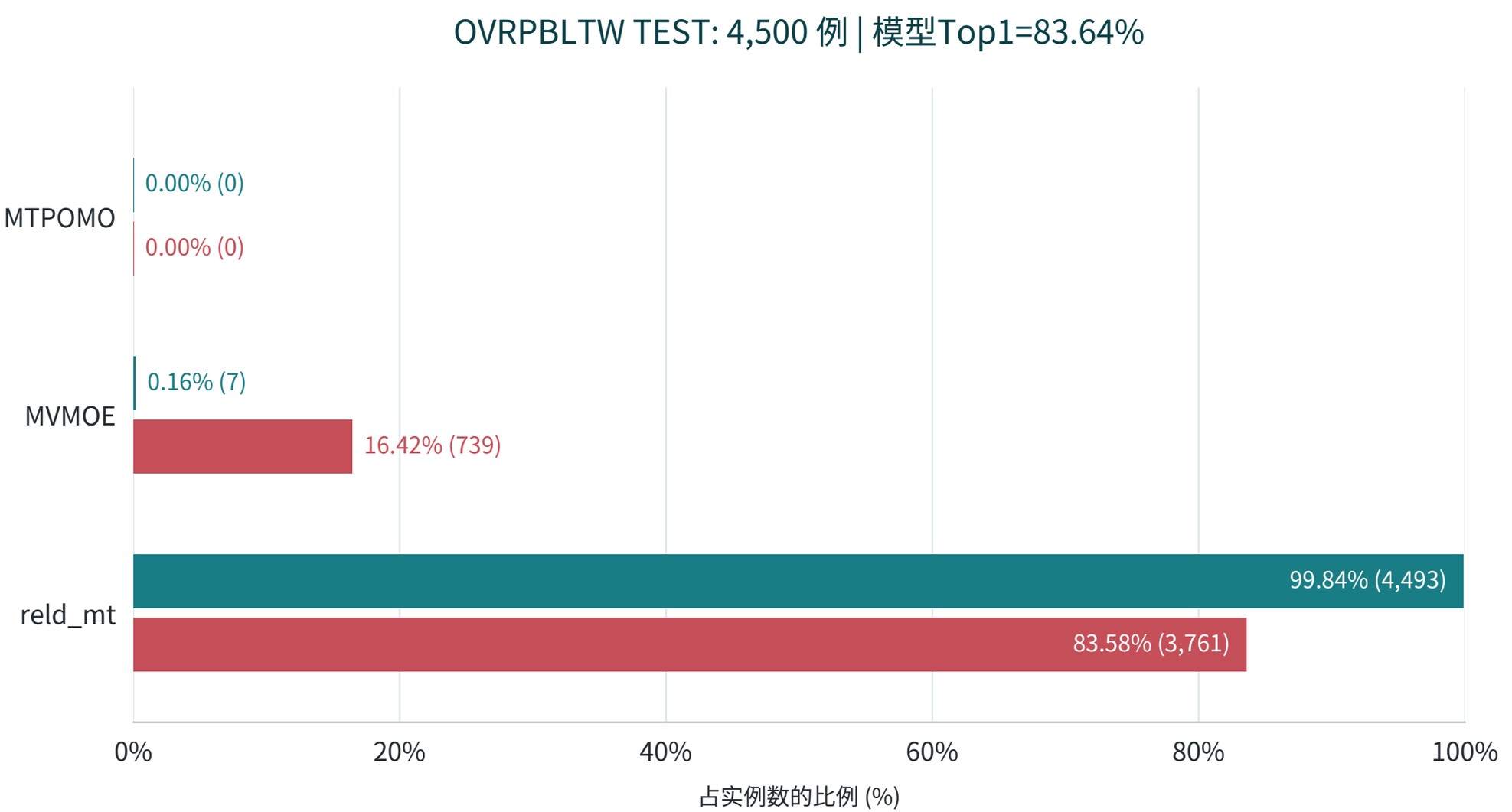
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